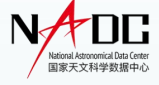


A photometric redshift catalogue of galaxies from DESI legacy imaging surveys DR10



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Introduction

- Two main methods for z_{phot} estimation:
 - Template-fitting:** Relies on SED templates, does not require training data, but accuracy depends on sufficient and unbiased templates.
 - Machine Learning:** Requires training data, achieves higher accuracy within the training range, but depends on sample representativeness and completeness.
- Goal:** To combine the two methods to improve z_{phot} estimation performance for DESI Legacy Imaging Survey DR10 galaxies.

Data

- Sample size: **2,417,805** entries.
- z_{spec} : Derived from 18 spectroscopic surveys, e.g. SDSS, DESI EDR, LAMOST.
- Photometry: Model and aperture magnitudes calculated from DESI LS DR10.

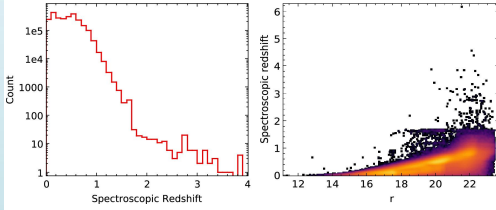


Fig1: Distribution of z_{spec} and r magnitude vs. z_{spec} .

Further Analysis

CatBoost with Adaptation to Different Subsamples.

Two-part Model

To analyze the Performance Differences Between Bright and Faint Samples.

Table 1. Performance Comparison Between Bright (KST_ML, $r \text{ mag} < 21.5$) and Faint (KST_MD, $r \text{ mag} > 21.5$) Samples, * indicates results after missing values are removed.

Sample	Bias	σ_{NMAD}	O(%)
KST_MD*	-2.15×10^{-3}	0.0274	2.84
KST_ML	1.14×10^{-3}	0.0129	0.42

Methods

EAZY and CatBoost are selected as the representative methods for template-fitting and machine learning, respectively.

EAZY: (Brammer et al. 2008)

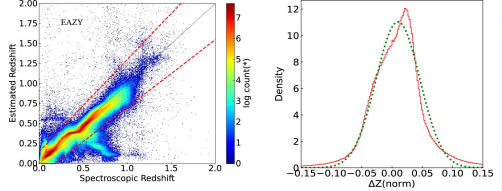


Fig2: scatter figure z_{phot} vs. z_{spec} and $\Delta z(\text{norm})$ for EAZY samples with $Q_z < 1$ (indicating accurate and reliable estimates).

CatBoost: (Dorogush et al. 2018)

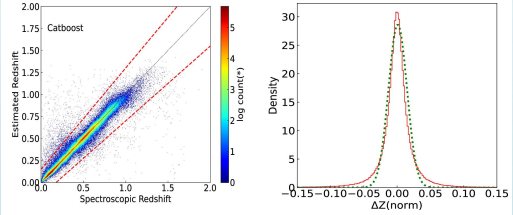


Fig3: scatter z_{phot} vs. z_{spec} and $\Delta z(\text{norm})$ for CatBoost samples (5-fold validation).

Two-step Model

To address the bias between low- and high- z distributions.

1st step: A classifier is trained to classify samples into low- z and high- z .

Binary Classification Acc: 99.26 %

2nd step: Separate regressors are trained and estimate z_{phot} within the low- and high- z subsamples based on classification results.

Table 2. Performance Comparison Between High redshift (KST_ZH, $z \geq 1$) and Low redshift (KST_ZL, $z < 1$) Samples.

Sample	Bias	σ_{NMAD}	O(%)
KST_ZH	5.3×10^{-4}	0.0249	1.50
KST_ZL	1.4×10^{-3}	0.0135	0.71

Application

z_{phot} estimation in DESI LS DR10 galaxies

- EAZY and one-step, two-part, two-step of CatBoost results are combined.
- 1,533,107,988 sources, 312,960,837 considered as high quality estimates.

Table 13. The estimated photometric redshifts of DESI DR10 galaxies. redshift, redshift_p, redshift_bp and redshift_m represent predicted redshifts by CatBoost models as shown in Figure 7. z_{EAZY} represents predicted redshifts by EAZY. Q_z demonstrates photometric redshift estimation quality with EAZY.

release	brickid	objid	RA	Dec	redshift	redshift_p	redshift_bp	redshift_m	z_{EAZY}	Q_z
10000	60641	10	0.426	-54.684	0.926	0.927	0.926	0.895	0.787	0.191
10000	60641	11	0.426	-54.828	0.836	0.954	0.836	0.903	1.125	0.296
10000	60641	31	0.429	-54.687	0.796	0.811	0.796	0.844	0.733	0.914
10000	60641	50	0.430	-54.710	0.853	0.912	0.853	0.969	0.960	0.198
10000	60641	52	0.430	-54.687	0.697	0.682	0.697	0.691	0.639	0.081
10000	60641	56	0.431	-54.683	0.823	0.980	0.823	1.102	1.078	0.446
10000	60641	62	0.431	-54.706	0.866	0.931	0.866	1.012	0.925	0.322
10000	60641	84	0.432	-54.747	0.830	0.821	0.830	0.819	0.680	0.162
10000	60641	90	0.432	-54.678	0.593	0.613	0.593	0.627	0.657	0.029
10000	60641	92	0.432	-54.650	0.380	0.853	0.380	0.824	2.026	0.376
10000	60641	125	0.433	-54.633	0.923	0.920	0.923	0.914	0.860	0.272
10000	60641	138	0.433	-54.693	0.858	0.895	0.858	0.902	0.766	0.065
10000	60641	142	0.434	-54.701	0.884	0.917	0.884	0.867	0.714	0.260
10000	60641	145	0.434	-54.732	0.790	0.784	0.790	0.796	0.730	0.495
10000	60641	150	0.434	-54.731	0.770	0.777	0.770	0.883	0.791	0.380
10000	60641	157	0.434	-54.625	0.641	0.648	0.641	0.657	0.620	0.648
10000	60641	165	0.434	-54.641	0.600	1.176	0.600	0.940	1.704	0.299
10000	60641	171	0.435	-54.868	0.397	0.314	0.397	0.459	4.309	0.032
10000	60641	172	0.435	-54.726	0.847	0.869	0.847	0.876	0.786	0.065
10000	60641	190	0.435	-54.812	0.830	0.875	0.830	0.834	0.752	0.117

Tab3: Examples of Final z_{phot} Estimate Files.

Future Work (in prep)

Emission Line Galaxies (ELG) Perform Worse Than BGS and LRG
→ Redshift Prediction for Specific Galaxy Types
Deep Learning can Integrate Image and Photometry Information
→ A Multimodal Photometric Redshift Estimation Model

Paper & Dataset Links

